Real-time assessment of listening effort using non-intrusive binaural prediction models

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Introduction

Prediction models of human speech perception can provide important insights into factors affecting listening effort. In practical applications, such models can be used, e.g., for algorithm development and optimization by complementing time- and cost-intensive listening tests.

For many such applications, however, there are critical challenges model approaches have to overcome in order to become applicable. For example, many intrusive (double-ended) models for prediction speech intelligibility (SI) or listening effort (LE) exist, but need (clean) reference signal, which is often not available. In contrast, non-intrusive metrics can operate based on (possibly degraded) speech signals alone, but are often confined to predicting speech quality, estimating signal-to-noise ratios (SNRs) as a proxy for SI or LE, or are rather specific for an individual use case (such as degraded telephone speech or naturalness of synthetic voices). Furthermore, some applications require online or even real-time processing, which limits the context available to the models to derive accurate predictions.

This contribution builds on the concept of Listening Effort prediction from Acoustic Parameters (LEAP), which computes listening effort predictions by using the output of a phoneme classifier commonly employed in automatic speech recognition [1, 2]. The model was validated in a number of studies in which model predictions were compared to subjective evaluation data, ranging from monitoring dialogs in broadcast signals, evaluating synthetic speech engines, speech enhancement algorithms, and different kinds of pathological speech, all of which had not been seen during training or model development. Current strengths and limitations of the model are discussed.

Listening Effort prediction from Acoustic Parameters (LEAP)

The basic approach of LEAP is to use *part* of an automatic speech recognition (ASR) engine, i.e., features are extracted from a single-channel audio stream, fed into a deep neural network, which computes probabilities for different phonemes for a given time frame (so-called phoneme posteriorgrams). However, rather than modeling the temporal sequence using language-specific models and deriving the most likely sequence of words or utterances from that (as typically done in ASR systems), perceptual predictions are directly derived from the posteriorgrams.

The underlying idea is to mathematically quantify the uncertainty of the ASR system, which is reflected in "smeared" phoneme posterior probabilities, in terms of a distance metric of vectors in the posteriorgrams called Mean Temporal Distance or M-Measure [3]. The ASR system used in the LEAP implementation of the present study was trained with approximately 8.000 hours of unprocessed speech material in a multi-condition setup using various types of noises, SNRs, talkers, and room impulse responses (for details, see [3]).

The design goal of the LEAP model is to predict categorical listening effort [4]. The scale contained fourteen categories from 1 effort scaling categorical unit (ESCU) corresponding to *no effort* to 13 ESCU (*extreme effort*) and also including a category of 14 ESCU (*only noise*), when listeners cannot detect any speech at all and, hence, a judgement of perceived LE cannot be reasonably made. To map the computed Mean Temporal Distance onto a scale reflecting perceived listening effort, subjective data from [5] were used, which consisted of in total 444 audio clips of about 10 s taken from German TV productions, respectively. The audio clips comprised various types of backgrounds and SNRs and had been presented to 20 normal-hearing subjects over open headphones (without video presentation). A comparison between the median ratings across listeners and the computed Mean Temporal Distance suggested a high correlation, which could be reasonably well approximated by a linear regression function. This function was then used to compute a mapping function from Mean Temporal Distance to ESCU, i.e., the same unit measured in subjective experiments using the categorical listening effort scale.

Figure 1 shows the resulting model predictions (abscissa) in comparison to the median LE ratings measured in the experiments (ordinate). Each data points corresponds to one of the 444 audio clips. The mapping function as well as all other model parameters including the training of the ASR system were then frozen, i.e., not adapted anymore to new, previously unseen audio data or new groups of listeners with which subjective LE assessments were collected.

In the following, predictions of the (frozen) model are compared to different experimental data sets collected in different contexts and with different (normal-hearing) listener panels.

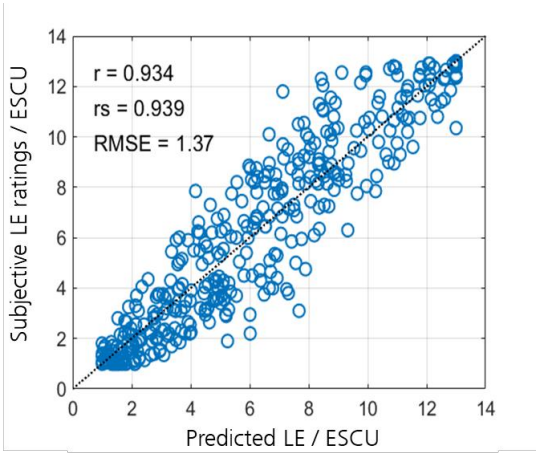


Figure 1: Median LE ratings measured in [5] for >400 audio clips of German broadcast material compared to predictions of LEAP after employing a linear mapping function from Mean Temporal Distance to the LE scale in ESCU. r: linear correlation coefficient, rs: rank correlation coefficient, RMSE: root-mean-square error.

Application examples

Figure 2 compares data of [4] compared to LEAP predictions. The data were collected with 15 normal-hearing subjects (mean data shown) using German sentences from the Oldenburg sentence test and stationary (OLNOISE) and time-varying speech-shaped noise (ICRA5-250) at different SNRs as commonly used in audiological research. High correlations and an RMSE in the order of 1 ESCU suggest high prediction accuracy for these type of masking noises.

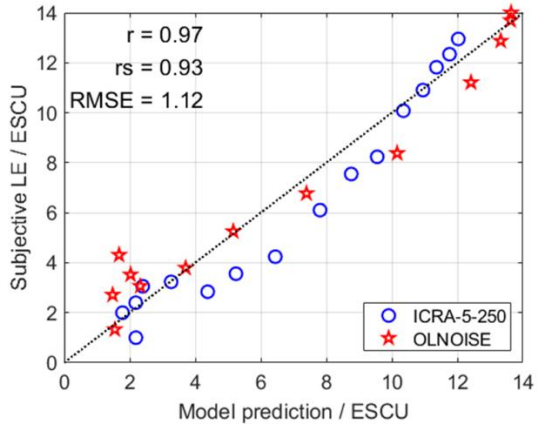


Figure 2: As Figure 1, but with subjective data from [4].

Figure 3 shows previously unpublished LE data collected with 16 normal-hearing subjects. German sentences from the Göttingen sentence test were used. The sentences were manipulated in different ways to produce different degrees of LE:

- Additive stationary speech-shaped noise as used above at SNRs from -10 to +12 dB SNR,
- Additive white noise at different SNRs -8 to +20 dB,
- Convolution with artificial room impulse responses with reverberation times from 1.3 to 12 s,
- Pitch manipulations using a Python toolbox from -10 to +6 semitones,
- Bandpass limiting with a fixed upper cutoff frequency of 8 kHz and varying lower cutoff frequencies,
- Bandpass limiting with a fixed lower cutoff frequency of 100 Hz and varying upper cutoff frequencies,
- Harmonic distortions ("overdrive") of various degrees using a Python toolbox.

For this data set, the correlations are somewhat lower and the RMSE is about twice as large as observed before. In general, however, the LEAP model predicts the detrimental effects of each individual speech degradation type with reasonably high accuracy (see correlation coefficients in the figure legend calculated separately), suggesting that the concept of using phoneme posteriorgrams as a basis for deriving LE predictions generalizes well to speech degradations not seen during the training of the ASR system.

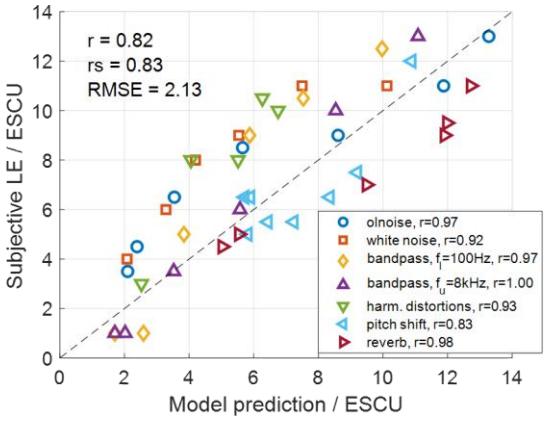


Figure 3: As Figure 1, but with subjective data collected in this study.

This is also true for musical noise degradations, which were assessed in terms of LE by 23 listeners using speech stimuli from the Oldenburg sentence test [6]. As illustrated in Figure 4, almost perfect correlation coefficients and an RMSE of about 1 ESCU were observed for this data set.

Similarly, high prediction accuracy was observed for the dataset reported in [7], see Figure 5. These data comprised data from eleven normal-hearing subjects collected with unprocessed speech from the same sentence test as well as speech processed by a so-called near-end listening enhancement algorithm which applies adaptive dynamic range compression and frequency-shaping to enhance speech perception in a given noisy environment, measured with two noise types (stationary speech-shaped noise and cafeteria noise) at different SNRs each. The high agreement of data and

predictions suggests that the LEAP concept also generalizes well to non-linear speech processing schemes not seen during the ASR training and model development.

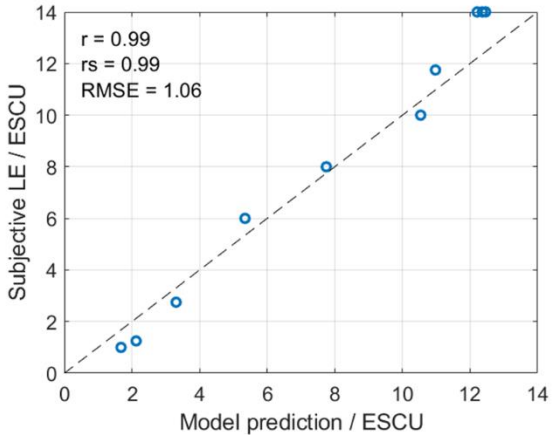


Figure 4: As Figure 1, but with subjective data from [6].

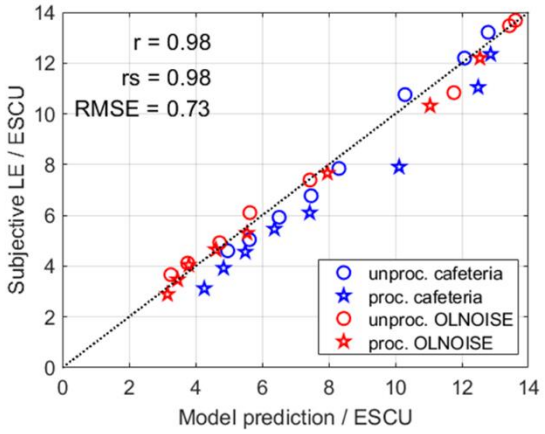


Figure 5: As Figure 1, but with subjective data from [7].

Another potential application for non-intrusive LE modeling is smart speaker output consisting of artificial voices, because these stimuli are typically created during runtime and reference signals are not available. Figure 6 shows data presented in [8], where 18 normal-hearing subjects assessed 91 audio clips (duration ~8-9 s) from different talkers, including synthetic voices. The speech had been recorded with a dummy head and mixed with various kinds of background noises and SNRs in different reverberant conditions (T_{60} from 0.2 to 2.3 s). The dichotic test signals were presented over headphones. For the model predictions, a simple better-ear selection was used. The high correlation and RMSE in the order of 1 ESCU suggest that this simplified approach is viable and that LEAP appears to provide good predictions also for synthetic speech.

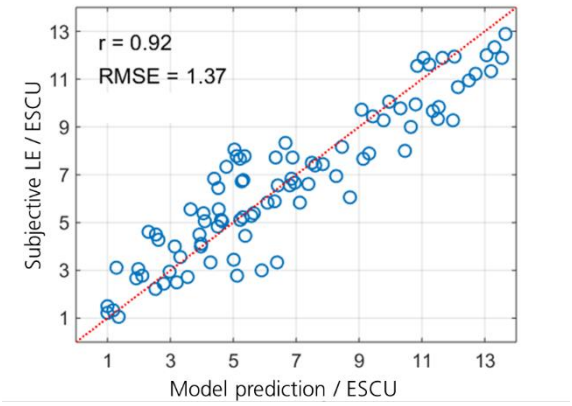


Figure 6: As Figure 1, but with subjective data from [8].

This was further investigated using a different dataset in [9]. Here, another group of 23 normal-hearing subjects assessed perceived LE of utterances form 20 synthetic voices with different degrees of naturalness. The voices were mixed with the same two types of noise at two SNRs each. Again, stimuli were presented via headphones and better-ear selection was used for modeling. The resulting data are shown in Figure 7. Different colors and symbols represent different combinations of noise type (blue: kitchen noise; red: office noise). Unlike in the other data sets, there was a systematic offset for the office noise, for which LEAP consistently predicted higher LE than rated by the listeners. Within each acoustic condition (SNR and noise type), there were again high correlations, indicating that the effect of synthetic voice quality (or naturalness) on LE was well predicted by LEAP.

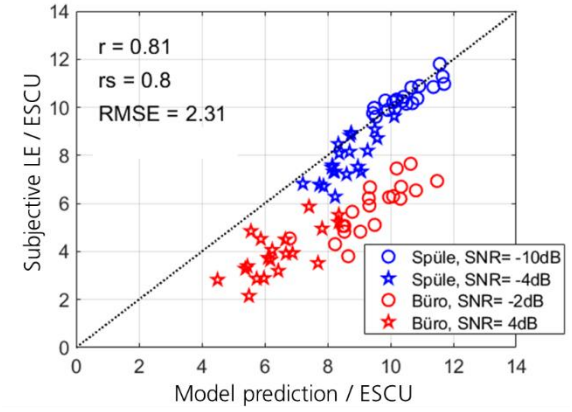


Figure 7: As Figure 1, but with subjective data from [9].

The same study conducted another experiment with the same 23 listeners. Here, two synthetic voices were mixed with five background sounds using a fixed (and different) SNR for each noise type. The stimuli were moderately reverberated and played back and recorded using different spatial configurations (see legend of Figure 8). As before, the dichotic test signals presented over headphones and better-ear modeling was applied.

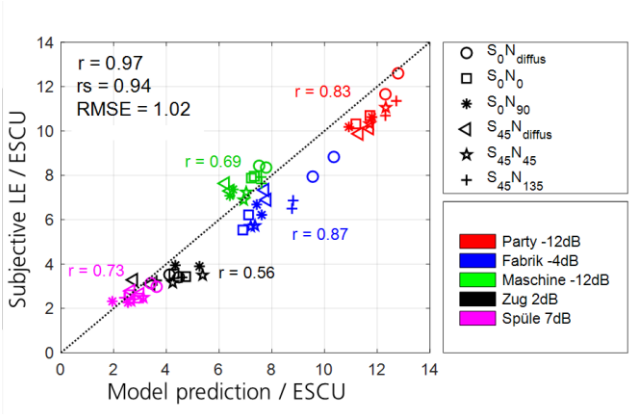


Figure 8: As Figure 1, but with subjective data from [9].

Taken all conditions together produced a similarly high prediction accuracy as observed above for the other datasets. Within each noise type, correlations were also significant but smaller. The smaller correlations are probably a result of the small variance for each noise type, while the significant correlations indicate that, again, the simple better-ear approach can be used to approximate the impact of different spatial configurations of speech and masker.

Summary and conclusions

This conference contribution highlights the potential of ASR-based posteriorgrams for predicting listening effort (LE) in a single-ended manner. Our findings demonstrate a high correlation with various subjective LE assessments, achieved without any task-specific training or model adaptations. The LEAP framework effectively accounts for the impacts of additive noise, reverberation, and different types of voice distortions and enhancements, making it a robust tool for LE evaluation.

Furthermore, the low-delay implementation of the model exists [10] which, combined with voice activity detection (VAD), is potentially well-suited for applications in several real-world scenarios. However, we acknowledge several limitations, including the underestimation of LE in scenarios with competing talkers, its applicability primarily to individuals with normal hearing (as far as the current evaluations are concerned), and its language dependency, with optimal performance observed in German and varying mapping functions for other languages [11]. In the present study, only a single-channel version of LEAP was considered with binaural processing coarsely approximated using a better-ear approach. Other studies have investigated non-intrusive models of binaural processing, which can be combined with "monaural" backends such as LEAP [10, 12]. Future work should address these limitations to broaden the applicability of this method across diverse listening environments and populations.

Acknowledgments

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